

PROBLEMSHEET-2
EPMTC301: MATHEMATICS FOR MACHINE LEARNING
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Here, we follow the same notations used in class. If you find any typos please let me know ASAP. It is highly recommended to try the first problem.

- (1) Do the following problems from the book **Schaum Series**:
- (a) Page number 176-193, Exercise number: 5.9, 5.11, 5.12, 5.14, 5.16, 5.18, 5.25, 5.26, 5.50, 5.61, 5.64, 5.65.
 - (b) Page number 205-224, Exercise number: 6.1, 6.8, 6.37, 6.40, 6.43.
- (2) Which of the following functions T from $\mathbb{R}^2$ to $\mathbb{R}^2$ are linear transformations?
- (a) $T(x_1, x_2) = (1 + x_1, x_2)$
 - (b) $T(x_1, x_2) = (x_2, x_1)$
 - (c) $T(x_1, x_2) = (x_1^2, x_2)$
 - (d) $T(x_1, x_2) = (\sin x_1, x_2)$
 - (e) $T(x_1, x_2) = (x_1 - x_2, 0)$
- (3) (a) Is there a linear transformation T from $\mathbb{R}^3$ into $\mathbb{R}^2$ such that $T(1, -1, 1) = (1, 0)$ and $T(1, 1, 1) = (0, 1)$? If yes, then how many are there?
- (b) Describe explicitly a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$ which has its range the subspace spanned by $(1, 0, -1)$ and $(1, 2, 2)$.
- (4) Let V be an n -dimensional vector space over the field F and let T be a linear transformation from V to V such that the range space and the null space of T are equal. Prove that n is even.
- (5) Let T be the linear operator on $\mathbb{C}^3$ for which

$$T(1, 0, 0) = (1, 0, i), \quad T(0, 1, 0) = (0, 1, 1), \quad T(0, 0, 1) = (i, 1, 0).$$

Is T invertible? If so, find its inverse.

- (6) Let T be the linear operator on $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1, x_1 - x_2, 2x_1 + x_2 + x_3).$$

Is T invertible? If so, find T^{-1} . Further show that T satisfies the equation $T^3 - 4T + 3I = 0$.

Just you can try the following problem.

- (7) (a) Find two linear operators T and U on $\mathbb{R}^2$ to $\mathbb{R}^2$ such that $TU = 0$ but $UT \neq 0$. Remark: Here 0 means the zero linear transformation.
- (b) Give an example of a linear transformation on $\mathbb{R}^2$ to $\mathbb{R}^2$ such that $T^2 = 0$ but $T \neq 0$.